Bill Of Materials - PDF for Variant [Maximal Configuration] of Project [Nixie Clock Avito x4 - SMD.PrjPcb] (No PCB Document Selected)

Component list

Project: Nixie Clock Avito x4 - SMD.PrjPcb
Variant: Maximal Configuration
Author: Dimasssius
Report Date: 04.06.2026

#	Designator	Comment	Description	Footprint	Quantity
1	ANT1	AN2051-245	Antenna, SMD, 2x5.1mm, 2, 0.5dBi, 110MHz, 2.45GHz, Antennas ROHS	ANTC2051	1
2	B1	CR1025	Battery Clip Coin, SMD, CR1025, Cell	CR1025	1
3	C1	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 250V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C1812_(M4532)	1
4	C2, C15, C17, C18, C20, C25, C26, C27	0.1uF	Capacitor, Ceramic (MLCC), SMD, 100nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	8
5	C3	22uF	Capacitor, Ceramic (MLCC), SMD, 22uF, 6.3V, ±20%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)	1
6	C4, C11	0.22uF	Capacitor, Ceramic (MLCC), SMD, 220nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	2
7	C5, C10	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 6.3V, ±10%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	2
8	C6, C14	6.8pF	Capacitor, Ceramic (MLCC), SMD, 6.8pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	2
9	C7, C9	0.1uF	Capacitor, Ceramic (MLCC), SMD, 100nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	2
10	C8	10uF	Capacitor, Ceramic (MLCC), SMD, 10uF, 6.3V, ±20%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	1
11	C12, C21	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 6.3V, ±10%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	2
12	C13	3pF	Capacitor, Ceramic (MLCC), SMD, 3pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	1
13	C16	2.4pF	Capacitor, Ceramic (MLCC), SMD, 2.4pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0402_(M1005)	1
14	C19, C23	100uF	Capacitor, Aluminum Electrolytic Capacitor, SMD, D5xL5.4mm, 100uF, 6.3V, ±20%, Aluminum Electrolytic Capacitors - SMD, ROHS	CAP_SMD_ELEC_ALUM_5MM_H5.4MM	2
15	C24	10nF	Capacitor, Ceramic (MLCC), SMD, 10nF, 50V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
16	D1	US1M	Diode, SMD, SMA(DO-214AC), 1kV, 1A, 75ns, 1.65V@1A, Independent Type, SMA, Fast Recovery / High Efficiency Diodes, ROHS	DO-214AC	1
17	D2	TL1838	Remote Receiver Sensor, 38.0kHz, 45m, Through Hole	TSOP4856	1
18	D3	LED0402	LED Indication - Discrete, 0402, 25mA, 249mcd, Colorless, Transparent Lens, -30℃~+85℃, Positive Stick, 516nm~531nm, Emerald, 120°, 80mW, 3V, LED Indication - Discrete, ROHS	LED_SMD_0402_(M1005)_GREEN	1
19	D4	SS12	Diode Schottky, SMD, SMA(DO-214AC), 20V, 1A, 500mV@1A, Schottky Diodes, ROHS	DO-214AC	1
20	D5, D6	LED0603	LED Indication - Discrete, 0603, 20mA, 225mcd, Colorless Transparent Lens, -20℃~+85℃, Positive Stick, 620nm~630nm, Red, 120°, 50mW, 2.3V, LED Indication - Discrete, ROHS	LED 0603_(1608)_0402A_(1005)	2
21	IC1, IC2, IC3, IC4, IC5	TLP291	Photovoltaic Output Optoisolator, SOP-4-1.27mm, 80V, 3.75kV, 50mA, 300mV@2.4mA, 8mA, 1, 5V, 1.25V, DC Transistor, Photovoltaic Output Optoisolators, ROHS	SOIC127P700X230-4N	5
22	IC6	74HC595D	Shift Register, SOIC-16, Serial To Serial Or Parallel Shift Register, ROHS	SOIC127P600X175-16N	1
23	IC7	W25Q64JVSSIQ	NOR FLASH, SOIC-8-208mil, 64Mbit, SPI, 133MHz, NOR FLASH, ROHS	SOIC127P798X216-8N	1
24	IC8	ESP8266EX	Microcontroller, QFN-32-EP(5x5), 72.2Mbps, 2400MHz~2483.5MHz, WiFi 2.4GHz, RF Transceiver ICs, ROHS	QFN50P500X500X90-33N-D	1
25	IC9	MAX668	AC-DC Controller and Regulator, MSOP-10, 100kHz~500kHz, 3V~28V, AC-DC Controllers and Regulators, ROHS	SOP50P490X110-10N	1
26	IC10	ATMEGA328P-AU	Microcontroller, TQFP-32(7x7), AVR, 20MHz, Microcontrollers (MCU/MPU/SOC), ROHS	QFP80P900X900X120-32N WITH SPI INTERFACE 2.54	1
27	IC11	AMS1117-3.3	Low Drop Out (LDO) Regulator, SOT-223, SOT-89-3, 72dB@(120Hz), 1A, Fixed 3.3V, Positive electrode, 16V, Voltage Regulators - Linear, Low Drop Out (LDO) Regulators, ROHS	SOT-89-3	1
28	IC13	RV-3028-C7	Real Time Clock, SMD3215-8P, I2C, Real Time Clocks, ROHS	SMD3215-8P	1
29	IC14	K155ID1	Decoder, DIP-16, 5V±5%, 25mA, 150ns, High Voltage Output, BCD to Decimal Decoders, RoHS	DIP16	1
30	J1, J3	Female Header 1x10-Pin	Female Header, 1x10-Pin, 10P, P=2.54mm, Direct Insert, H:8.5mm, W:2.54mm, D:3.2mm, Single Row, Black Brass, Top Square, Hole Plugin, 3A, 20mΩ, 1000V, -40℃~+105℃, Female Headers, ROHS	HEADER_FEMALE_10PIN	2
31	J2	USB_C	USB Connector, 5A, USB 3.1, 1 Horizontal Attachment, 16P, Female, -25℃~+85℃, Type-C, SMD, USB Connectors, ROHS	USB_C_DOUBLE	1
32	L1	LQG15HS2N2B02D	Fixed Inductor, SMD, 0402, 2.2nH, 900mA, ±5%, 90mΩ, Fixed Inductors, RoHS	IND_SMD_0402_(M1005)	1
33	LS1	FUET-4020	Buzzer, SMD, 4x4mm, 65dB, Passive (Externally Driven), Electromagnetic, 4kHz, Buzzers ROHS	FUET-4020	1
34	Q1, Q2, Q4	MMBTA42	Transistor Bipolar (BJT), NPN, SOT-23, 300V, 225mW, 25@10mA, 10V, 500mA, NPN Bipolar (BJT), ROHS	SOT105P255X140-3N	3
35	Q3	AON7264E	MOSFET (N-Channel), DFN-8(3x3), 60V, 28A, 11W, 13.3mΩ@4.5V, 28A, 1.8V, 1 N-channel, MOSFETs, ROHS	DFN-8-EP(3x3)	1
36	R1, R4, R6, R7, R9, R24, R25, R27, R31	10kΩ	Resistor, Thick Film Resistor, SMD, 10kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	9
37	R2, R8	33kΩ	Resistor, Thick Film Resistor, SMD, 33kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	2
38	R3	470Ω	Resistor, Thick Film Resistor, SMD, 470Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
39	R5	56kΩ	Resistor, Thick Film Resistor, SMD, 56kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
40	R10	3kΩ	Resistor, Thick Film Resistor, SMD, 3kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
41	R11	220Ω	Resistor, Thick Film Resistor, SMD, 220Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0402_(M1005)	1
42	R12	0.1Ω	Resistor, Thick Film Resistor, SMD, 0.1Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R1206_(M3216)	1
43	R13	1kΩ	Resistor, Thick Film Resistor, SMD, 1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0402_(M1005)	1
44	R14	1.5MΩ	Resistor, Thick Film Resistor, SMD, 1.5MΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
45	R15	499Ω	Resistor, Thick Film Resistor, SMD, 499Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0402_(M1005)	1
46	R16	12kΩ	Resistor, Thick Film Resistor, SMD, 12kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0402_(M1005)	1
47	R17	8.2kΩ	Resistor, Thick Film Resistor, SMD, 8.2kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
48	R18, R21	10kΩ	Resistor, Thick Film Resistor, SMD, 10kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0402_(M1005)	2
49	R19	3kΩ	Resistor, Potentiometer, SMD, SMD-3P, 3.9x3.1mm, Potentiometer, 3kΩ, ±25%, 150mW, Potentiometers, Variable Resistors, ROHS	TC33X	1
50	R20	499kΩ	Resistor, Thick Film Resistor, SMD, 499kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
51	R22	200Ω	Resistor, Thick Film Resistor, SMD, 200Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
52	R23	100Ω	Resistor, Thick Film Resistor, SMD, 100Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
53	R26	GL5528	Photoresistor, Through Hole, P=3mm, 10kΩ~20kΩ, 540nm, 1MΩ, Photoresistors, ROHS	Photoresistor	1
54	R28	5.1kΩ	Resistor, Thick Film Resistor, SMD, 5.1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
55	R29, R30, R32	4.7kΩ	Resistor, Thick Film Resistor, SMD, 4.7kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	3
56	R34, R35, R36, R37	1kΩ	Resistor, Thick Film Resistor, SMD, 1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	4
57	S1, S2, S3, S4	TS-1009B	Switch Tactile Through Hole, 4.5x4.5mm, Round Button, 50mA, Bend-in 4.5mm, SPST, 12V, Plugin, Tactile Switches, ROHS	TS-1009B	4
58	S5	TSA016A2518C	Switch Tactile SMD, 3.35x2.5mm, Oval button, 50mA, Standing paste, 4.2mm ,180gf, 12V, SMD Tactile Switches, ROHS	TS20	1
59	S6	BME280	Development Board, LGA-8(2.5x2.5), Temperature and Humidity Sensor, ROHS	BME280_DEVBOARD	1
60	S7, S8, S9, S10	TS-1187A-B-A-B	Switch Tactile SMD, 5.1x5.1mm, Without bracket, 50mA, 100000 Times, 160gf, 12V, Round, SMD Tactile Switches, ROHS	TS-1187A-B-A-B	4
61	TR1	EE0504S	Power Transformer, SMD-5P, 1:10, 6.4x6.4mm, Power Transformers, ROHS	EE0504S	1
62	U1	CH340C	USB Converter, SOP-16, 2Mbps Transceiver, USB 2.0, USB Converters, ROHS	SOIC127P600X180-16N	1
63	Y1	XTM20026000FT00351005	Crystal Oscillator, SMD, SMD3225-4P, 3.2x2.5mm, 26MHz, 12pF, ±10ppm, ±20ppm, Crystals, ROHS	XTL_4_3225	1
64	Y2	CSTCE16M0V53-R0	Ceramic Resonators, SMD, SMD3213-3P, 3.2x1.3mm,16MHz, 15pF, 40Ω, Ceramic Resonators, ROHS	XTL_3_3213	1
Notes					**106**